

Dihedral 1

Arham Lodha

September 15, 2023

Problem 1. Let $\langle, \rangle$ be the standard dot product on $\mathbb{R}^n$. Let $O(n)$ be the subset of matrices in GL_n such that $\langle AX, AY \rangle = \langle X, Y \rangle$. Show this is a subgroup, and that A lies in $O(n)$ iff $AA^T = I$, where T means transpose (note $\langle X, Y \rangle = X^t Y$, if X and Y are column vectors).

Claim 1A. $\forall A \in GL_n(\mathbb{R})$, if $A^T A = I \iff A \in O(n)$

Proof. ($\implies$): Suppose for $A \in GL_n(\mathbb{R})$, $A^T A = I$. Then $\forall x, y \in \mathbb{R}^n$ then $\langle Ax, Ay \rangle = (Ax)^T Ay = x^T A^T Ay = x^T Iy = x^T y = \langle x, y \rangle$. Thus $A \in O(n)$,

($\impliedby$): Suppose $A \in O(n)$. Then $\forall x, y \in \mathbb{R}^n$ then $\langle Ax, Ay \rangle = \langle x, y \rangle$, $\langle Ax, Ay \rangle = (Ax)^T Ay = x^t A^t Ay$ and $\langle x, y \rangle = x^t y$. Thus $x^t A^t Ay = x^t y$. Since this holds for all x and y let $x = e_i^n$ and e_j^n for $i, j \in [1, n]$ where e_k^n is the k th standard basis. $Ae_i^n = c_i$ or the i th column of A . $\langle e_i^n, e_j^n \rangle = \delta(i, j)$. Thus $\langle Ae_i^n, Ae_j^n \rangle = (Ae_i^n)^T Ae_j^n = c_i^T c_j = \delta(i, j)$. Let $A = [c_1 \ \cdots \ c_n]$ where c_i is the i th column vector.

$$A^T A = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \begin{bmatrix} c_1 & \cdots & c_n \end{bmatrix} = \begin{bmatrix} c_1^T c_1 & \cdots & c_1^T c_n \\ \vdots & \ddots & \vdots \\ c_n^T c_1 & \cdots & c_n^T c_n \end{bmatrix} = \begin{bmatrix} \delta(1,1) & \cdots & \delta(1,n) \\ \vdots & \ddots & \vdots \\ \delta(n,1) & \cdots & \delta(n,n) \end{bmatrix} = \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix} = I.$$

Thus $A^T A = I$. □

Claim 1B. $O(n)$ is a subgroup of $GL_n(\mathbb{R})$

Proof. By definition, $O(n) \subset GL_n(\mathbb{R})$.

1. **Closure:** $\forall A, B \in O(n)$, $\forall x, y \in \mathbb{R}^n$, $\langle ABx, ABy \rangle = \langle Ax, Bx \rangle = \langle x, y \rangle$ by definition of A and B . Thus $AB \in O(n)$.
2. **Identity:** $I_n \in O(n)$, because $\forall x, y \in \mathbb{R}^n$ $I_n x = x$ and $I_n y = y$ thus $\langle I_n x, I_n y \rangle = \langle x, y \rangle$
3. **Inverses:** $\forall A \in O(n)$, $\forall x, y \in \mathbb{R}^n$ $\langle A^{-1}x, A^{-1}y \rangle = (A^{-1}x)^T A^{-1}y$. By the previous proof we know $A^T A = I$, thus $A^{-1} = A^T$. Thus $x^t (A^{-1})^T A^{-1}y = x^t A A^{-1}y = x^t y = \langle x, y \rangle$

$O(n)$ is a subgroup of $GL_n(\mathbb{R})$. □

Problem 2. Let $\langle, \rangle$ be the standard dot product on $\mathbb{C}^n$ where $\langle X, Y \rangle$ is $X^t \bar{Y}$, where $\bar{}$ means take complex conjugate of all the coordinates. We let $U(n)$ be the subgroup of GL_n that preserve this, show A lies in here iff $A(\bar{A})^t = I$.

Claim 2A. $\forall A \in GL_n(\mathbb{C})$ if $A(\bar{A})^t = I \iff A \in U(n)$

Proof. ($\implies$): Suppose $A \in GL_n(\mathbb{R})$ and $A(\bar{A})^T = I$. Then $\forall x, y$

$$\begin{aligned}\langle Ax, Ay \rangle &= (Ax)^T (\bar{A}y) \\ &= x^T A^T \bar{A}y \\ &= x^T Iy \\ &= x^T \bar{y} \\ &= \langle x, y \rangle\end{aligned}$$

Thus by definition $A \in U(n)$

($\impliedby$): Suppose $A \in U(n)$. Then $\forall x, y$

$$\begin{aligned}\langle Ax, Ay \rangle &= \langle x, y \rangle \\ (Ax)^T (\bar{A}y) &= x^t \bar{y} \\ x^T A^T \bar{A}y &= x^t \bar{y}\end{aligned}$$

Since this holds for all x and y let $x = e_i^n$ and $y = e_j^n$ for $i, j \in [1, n]$ where e_k^n is the k th standard basis. $Ae_i^n = c_i$ or the i th column of A . $\langle e_i^n, e_j^n \rangle = \delta(i, j)$. Thus

$$\begin{aligned}\langle Ae_i^n, Ae_j^n \rangle &= \langle e_i^n, e_j^n \rangle \\ (Ae_i^n)^T (Ae_j^n) &= \delta(i, j) \\ (c_i)^T \bar{c}_j &= \delta(i, j)\end{aligned}$$

Let $A = [c_1 \ \cdots \ c_n]$ where c_i is the i th column vector. $A^T A = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} [\bar{c}_1 \ \cdots \ \bar{c}_n] = \begin{bmatrix} c_1^T \bar{c}_1 & \cdots & c_1^T \bar{c}_n \\ \vdots & \ddots & \vdots \\ c_n^T \bar{c}_1 & \cdots & c_n^T \bar{c}_n \end{bmatrix} =$

$$\begin{bmatrix} \delta(1,1) & \cdots & \delta(1,n) \\ \vdots & \ddots & \vdots \\ \delta(n,1) & \cdots & \delta(n,n) \end{bmatrix} = \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix} = I. \text{ Thus } A^T \bar{A} = I. \quad \square$$

Claim 2b. $U(n)$ is a subgroup of $GL_n(\mathbb{C})$

Proof. By definition $U(n) \subset GL_n(\mathbb{C})$.

1. **Closure:** $\forall A, B \in U(n), \forall x, y \in \mathbb{R}^n, \langle ABx, AB y \rangle = \langle Ax, Ax \rangle = \langle x, y \rangle$ by definition of A and B . Thus $AB \in U(n)$.
2. **Identity:** $I_n \in U(n)$, because $\forall x, y \in \mathbb{R}^n, I_n x = x$ and $I_n y = y$ thus $\langle I_n x, I_n y \rangle = \langle x, y \rangle$
3. **Inverses:** $\forall A \in O(n), \forall x, y \in \mathbb{R}^n, \langle A^{-1}x, A^{-1}y \rangle = (A^{-1}x)^T A^{-1}y$. By the previous proof we know $A^T \bar{A} = I$, thus $\bar{A}^{-1} = A^T$. Thus $x^t (\bar{A}^{-1})^T \bar{A}^{-1}y = x^t \bar{A} \bar{A}^{-1}y = x^t y = \langle x, y \rangle$

$\square$

Problem 3. Let A be in $U(n)$ explain why all the eigenvalues have complex norm one. Conclude that in $O(n)$, the $\det$ is either 1 or -1.

Claim 3a. Let A be in $U(n)$ then all its eigenvalues have complex norm one.

Proof. Let λ be a eigenvalue of A with associated eigenvector v . Thus $Av = \lambda v$.

$$\begin{aligned}\langle Av, Av \rangle &= \langle v, v \rangle \\ (Av)^T (\bar{A}v) &= v^T \bar{v} \\ (\lambda v)^T (\bar{\lambda}v) &= v^T \bar{v} \\ \lambda \bar{\lambda} v^T \bar{v} &= v^T \bar{v}\end{aligned}$$

Since v is a eigenvector, $v \neq 0$. Thus $\lambda \bar{\lambda} = 1 = |\lambda|^2$ because $\lambda \bar{\lambda} v^T \bar{v} = v^T \bar{v}$ is true. Thus the complex norm of $\lambda = 1$ □

Claim 3b. $A \in O(n) \implies \det A = \pm 1$

Proof. Suppose $A \in O(n)$. By Claim 1A, $A^T A = I$. By properties of the determinant, $\det(A^T) = \det(A)$. $\det(I) = 1 = \det(A) \det(A^T) = \det(A)^2$. Thus $\det(A) = \pm 1$ □

Claim 4. $A \in O(n)$, show that its distinct eigenspaces are orthogonal

Proof. Let $A \in O(n)$. Let λ_1, λ_2 be the eigenvalues whose respective eigenvectors are v, w . Since $A \in O(n)$, $\langle Av, Aw \rangle = \langle v, w \rangle$. Thus $\langle \lambda_1 v, \lambda_2 w \rangle = \lambda_1 \bar{\lambda}_2 \langle v, w \rangle = \langle v, w \rangle$ Thus $(\lambda_1 \bar{\lambda}_2 - 1) \langle v, w \rangle = 0$. If $\lambda_1 \bar{\lambda}_2 - 1 = 0$, then $\lambda_1 \bar{\lambda}_2 = 1$, $\bar{\lambda}_2 = \lambda_1^{-1}$. Since $|\lambda_1|^2 = 1 = \lambda_1 \bar{\lambda}_1$. Thus $\bar{\lambda}_2 = \bar{\lambda}_1$. Thus $\lambda_1 = \lambda_2$. Thus v and w are in the same eigenspace.

However if $\lambda_1 \bar{\lambda}_2 - 1 \neq 0$, then $\lambda_1 \neq \lambda_2$ and $\langle v, w \rangle = 0$. Thus vectors in distinct eigenspaces are orthogonal and the eigenspaces are orthogonal. □

Problem 5. Now we think about $O(2)$. We have $\det: O(2) \rightarrow \{+1, -1\}$ by 3). The kernel is called $SO(2)$. Let A be in $\det^{-1}(-1)$ – so the other coset of $SO(2)$ (note $SO(2)$ has index 2). Prove that its eigenvalues are 1 and -1. Now using 4 explain why A is a reflection. Note A is completely determined by its eigenspace $V_1(A)$ – which is the line of reflection.

Claim 5a. $A \in \det^*(-1)$, eigenvalues are 1 and -1.

Proof. Let $A \in \det^*(-1)$. Assume the contrary λ_1 and λ_2 not be real. $P_A(\lambda)$ will have either 2 real roots or two complex roots which are conjugates. Thus $\lambda_1 = \lambda$ and $\lambda_2 = \bar{\lambda}$. Since $-1 = \lambda \bar{\lambda} = |\lambda|$. But complex norm is always positive. This forms a contradiction thus the eigenvalues of A must be real. Furthermore if the eigenvector of λ_1 is v , $\langle Av, Av \rangle = \lambda_1^2 \langle v, v \rangle = \langle v, v \rangle$ thus $\lambda_1^2 = 1$. Similarly $\lambda_2^2 = 1$. Thus $\lambda_1 \lambda_2 = -1$ for real λ thus the eigenvalues must be 1 and -1. □

Claim 5b. Now using 4 explain why A is a reflection. Note A is completely determined by its eigenspace $V_1(A)$ – which is the line of reflection.

Proof. By 4b, $V_1(A) \perp V_{-1}(A)$. Moreover since A has 2 distinct eigenvalues, it is diagonalizable, thus $\dim V_1(A) + \dim V_{-1}(A) = \dim A$. Let $v \in V_1(A)$ s.t $v \neq 0$, by definition $Av = v$, thus $V_1(A)$ is the subspace formed by $v \neq 0$ and is thus a line and all vectors on the line are fixed by A . Furthermore, $w \in V_{-1}(A)$ s.t $w \neq 0$, by definition $Aw = -w$, and thus $V_{-1}(A)$ is the subspace formed by $w \neq 0$ and is thus a line and all vectors on the line are flipped by A . Further more for $x \in \mathbb{R}^2$, $x = av + bw$ and $Ax = aAv + bAw = av - bw$, thus the component of $x \parallel v$ is fixed and component of $x \perp v$ is flipped. Thus A is a reflection □

Problem 6. *Orientation: Note that for a non-zero vector in $\mathbb{R}^2$, there are two orthogonal unit vectors, one on the left, one on the right (and they are negatives of each other). For a matrix $A \in O(2)$ explain why the two columns are orthogonal unit vectors. Show $A \in SO(2)$ iff c_2 is on the left of c_1 (c_i being the i th column). Show that rotation by some fixed angle is linear (don't use coordinates, use the "parallelogram" definition of vector addition), and lies in $SO(2)$. Show $SO(2)$ is exactly the set of rotations, and thus isomorphic to $U(1)$, the unit circle, as a group.*

Claim 6a. $A \in O(n) \implies A$ is orthonormal

Proof. By a part of the implied by section of claim 1A, I have proven, $\langle c_i, c_j \rangle = \delta(i, j)$ if $A \in O(n)$. Thus $\langle c_i, c_i \rangle = 1 = |c_i|^2$ and $|c_i| = 1$ and $\langle c_i, c_j \rangle = 0$ if $i \neq j$. Thus $i \neq j$, $c_i \perp c_j$. Thus every column of A has a norm of 1 and is orthogonal to all the other columns and A is orthonormal by definition. □

Thus $A \in O(2)$ is orthonormal.

Claim 6b. $A \in SO(2) \iff c_2$ is on the left of c_1 (c_i being the i th column).

Proof. Let the first column of A be $c_1 = \begin{bmatrix} a \\ b \end{bmatrix}$ since A is in $SO(2)$ its orthonormal so $|c_1| = 1a^2 + b^2 = 1$. $c_1 = Ae_1$ and $c_2 = Ae_2$ by definition of matrix multiplication. Since A preserves the inner product, $c_1 \perp c_2$, because $\langle Ae_1, Ae_2 \rangle = \langle c_1, c_2 \rangle = \langle e_1, e_2 \rangle = 0$. Thus there only two possibilities for $c_2 = \begin{bmatrix} -b \\ a \end{bmatrix} = w$ or $c_2 = \begin{bmatrix} b \\ -a \end{bmatrix} = w$. If $c_2 = w$, then $A = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$ and $\det(A) = -(a^2 + b^2) = -1$ so $A \notin SO(2)$ which is a contradiction. Thus $c_2 = v$ which is on the left of c_1 and A is $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ and $\det(A) = 1$ and $A \in SO(2)$. □

Claim 6c. *Rotation by a fixed angle is linear*

Proof. Let R_x be the rotation operator by x radians. Rotation preserves angles and lengths.

Let $OABC$ be a parallelogram with a corner at the origin. Let OA and OC be the spanning vectors of the parallelogram. $OB = OA + OC$ by definition of vector addition. $R_x(OA)$ and $R_x(OC)$ are rotations of x radians of OA and OC and since x is fixed $R_x(OA)$ and $R_x(OC)$ are rotated by the same angle. Furthermore $|OA| = |R_x(OA)|$ and $|OC| = |R_x(OC)|$. Furthermore the angle between OA and OC stays the same after the rotation because they are rotated by the same angle. $R_x(OA) + R_x(OC)$ form the same parallelogram because the angles and lengths of sides have not changed, Thus $R_x(OA) + R_x(OC) = R_x(OB)$.

Lets consider a vector v and $R_x(v)$ is the result of rotating v . When we scale the v by a , we are simply scaling the vector and not changing the angle between it and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and thus after the rotation

the length of v doesn't change just the angle of rotation. Thus first rotating v and then scaling it will lead to the same result. $R_x(av) = aR_x(v)$

Alternate Proof: $R_x = \begin{bmatrix} \cos(x) & -\sin(x) \\ \sin(x) & \cos(x) \end{bmatrix}$ takes $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} \cos(x) \\ \sin(x) \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} -\sin(x) \\ \cos(x) \end{bmatrix}$.

$\det R_x = \cos^2(x) + \sin^2(x) = 1$ and $R_x^T R_x = \begin{bmatrix} \cos^2(x) + \sin^2(x) & -\cos(x)\sin(x) + \cos(x)\sin(x) \\ \cos(x)\sin(x) - \cos(x)\sin(x) & \cos^2(x) + \sin^2(x) \end{bmatrix} =$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$. Thus by claim 1A, R_x is in $O(2)$. By problem 5, $R_x \in SO(2)$. Furthermore thus for all $x \in \mathbb{R}$, $\langle Ax, Ax \rangle = \langle x, x \rangle = |x|^2$, thus applying A doesn't change the length of x . Thus R_x is rotation by x degrees. Since R_x is a matrix, rotation by a fixed degree x is linear. $\square$

Claim 6d. $\forall A \in SO(2)$, $A = R_x$ for some x .

Proof. $\forall A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SO(2)$. Since A is orthonormal, $a^2 + c^2 = 1$, $b^2 + d^2 = 1$, $ab + cd = 0$.

Furthermore since $A \in SO(2)$, $\det A = ad - bc = 1$. $a^2 = 1 - c^2$ thus $a \leq 1$. Let $x \in \mathbb{R}$, $a = \cos(x)$. Thus $c^2 = 1 - \cos^2(x) = \sin^2(x)$ by Pythagorean Identity and $c = \sin(x)$. $ab + cd = \cos(x)b + \sin(x)d = 0$

and $ad - cb = -\sin(x)b + \cos(x)d = 0$. Thus $\begin{bmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{bmatrix} \begin{bmatrix} b \\ d \end{bmatrix} = R_x^T \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ because

$\begin{bmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{bmatrix} = R_x^T$ by definition. $\begin{bmatrix} b \\ d \end{bmatrix} = R_x \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(x) & -\sin(x) \\ \sin(x) & \cos(x) \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin(x) \\ \cos(x) \end{bmatrix}$. $b =$

$-\sin(x)$ and $c = \cos(x)$. Thus $A = \begin{bmatrix} \cos(x) & -\sin(x) \\ \sin(x) & \cos(x) \end{bmatrix} = R_x$. Thus all matrices in $SO(2)$ are rotation matrices. $\square$

Claim 6e. $SO(2)$ iso to $U(1)$

Proof. By Claim 6d, $\forall A \in SO(2)$, $A = R_x$ for $x \in \mathbb{R}$ where R_x is the rotation matrix by fixed angle x . Let $f : SO(2) \rightarrow U(1)$ such that it takes $R_x \rightarrow [e^{ix}]$. Let $g : U(1) \rightarrow SO(2)$ such that it takes $[e^{ix}] \rightarrow R_x$. Let $R_x \in SO(2)$, $g(f(R_x)) = g([e^{ix}]) = R_x$. $\forall u \in U(1)$, $u = [e^{ix}]$ for some $x \in \mathbb{R}$, thus $f(g(u)) = f(g([e^{ix}])) = f(R_x) = [e^{ix}] = u$. Thus f and g are inverses and f is bijective.

$R_x R_y = \begin{bmatrix} \cos(x)\cos(y) - \sin(x)\sin(y) & -\cos(x)\sin(y) - \sin(x)\cos(y) \\ \sin(x)\cos(y) + \cos(x)\sin(y) & \cos(x)\sin(y) + \sin(x)\cos(y) \end{bmatrix} = \begin{bmatrix} \cos(x+y) & -\sin(x+y) \\ \sin(x+y) & \cos(x+y) \end{bmatrix} =$

R_{x+y} . Thus $f(R_x R_y) = f(R_{x+y}) = [e^{i(x+y)}] = [e^{ix}][e^{iy}] = f(R_x)f(R_y)$. Thus f is a homomorphism and is an isomorphism. Thus $SO(2)$ iso to $U(1)$. $\square$

Claim 7a. Conjugation in $O(2)$: Let r_L be reflection in the line L , and g any element of $O(2)$. Explain why $gr_L g^{-1}$ (the conjugate) is again reflection (hint, think determinant).

Proof. Because r_L is a reflection, $r_L \in \det^*(-1)$ and $\det(r_L) = -1$. If $g \in O(2)$, $\det(g) = \pm 1$ by claim 3B. Since $g \in O(2)$, $g^{-1} = g^T$, and $\det g^T = \det(g)$. Thus $\det(gr_L g^{-1}) = \det(g)\det(r_L)\det(g^{-1}) = \det(r_L)\det(g)\det(g)$ because $\mathbb{R}$ is abelian wrt multiplication. $\det(gr_L g^{-1}) = -1(\pm 1)^2 = -1(1) = -1$. Since $O(2)$ is a group, $gr_L g^{-1} \in O(2)$ and since $\det(gr_L g^{-1}) = -1$, $gr_L g^{-1}$ is a reflection. $\square$

Claim 7b. Each reflection is just a conjugation of reflection across the x axis by rotation

Proof. By 7a, we know gr_xg^{-1} is a reflection for all $g \in O(2)$. Let $g = R_x$ or a rotation about a fixed angle $x \neq 0$. If $x = 0$, then L is the line generated by $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

$$R_x r_x R_x^{-1} = R_x r_x R_x^T \begin{bmatrix} \cos(x) & -\sin(x) \\ \sin(x) & \cos(x) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{bmatrix} = \begin{bmatrix} \cos(2x) & \sin(2x) \\ \sin(2x) & -\cos(2x) \end{bmatrix}$$

Thus the matrix $\begin{bmatrix} \cos(2x) & \sin(2x) \\ \sin(2x) & -\cos(2x) \end{bmatrix}$ is a reflection.

$$R_x r_x R_x^T - I = \begin{bmatrix} \cos(2x) - 1 & \sin(2x) \\ \sin(2x) & -(\cos(2x) + 1) \end{bmatrix}$$

Eigenvector with eigenvalue 1 is $v = \begin{bmatrix} -\sin(2x) \\ \cos(2x) - 1 \end{bmatrix}$ where $x \in \mathbb{R}$. $\forall x \in \mathbb{R}$, v generates all possible lines of reflection.

Thus every reflection is just a conjugation of the reflection over the x axis. $\square$

Claim 7c. *The reflection line of gr_Lg^{-1} is $g(L)$*

Proof. For every $v \in g(L)$, $g^{-1}v = w$ where $w \in V_1(r_L) = L$. Thus $r_Lg^{-1}v = r_Lw = w$. Then $g(w) = v$. $v \in V_1(gr_Lg^{-1})$. Thus the reflection line is $g(L)$. $\square$

Claim 7d. *Rotations commute*

Proof. All of $SO(2)$ is a rotation. $\forall R_x, R_y \in SO(2)$. $R_x R_y = \begin{bmatrix} \cos(x)\cos(y) - \sin(x)\sin(y) & -\cos(x)\sin(y) - \sin(x)\cos(y) \\ \sin(x)\cos(y) + \cos(x)\sin(y) & \cos(x)\sin(y) + \sin(x)\cos(y) \end{bmatrix}$
 $\begin{bmatrix} \cos(x+y) & -\sin(x+y) \\ \sin(x+y) & \cos(x+y) \end{bmatrix} = R_{x+y}$ and $R_y R_x = \begin{bmatrix} \cos(x)\cos(y) - \sin(x)\sin(y) & -\cos(x)\sin(y) - \sin(x)\cos(y) \\ \sin(x)\cos(y) + \cos(x)\sin(y) & \cos(x)\sin(y) + \sin(x)\cos(y) \end{bmatrix} =$
 $\begin{bmatrix} \cos(x+y) & -\sin(x+y) \\ \sin(x+y) & \cos(x+y) \end{bmatrix} = R_{x+y}$. Thus $R_x R_y = R_y R_x$ and thus $SO(2)$ is abelian. $\square$

Claim 7d. *If g is a rotation $gR_\theta g^{-1} = R_\theta$*

Proof. Since rotations commute by 7d, $gR_\theta g^{-1} = gg^{-1}R_\theta = R_\theta$. $\square$

Claim 7e. *If g is a reflection $gR_\theta g^{-1} = R_{-\theta}$*

Proof. Let $g \in \det^*(-1)$ and $g \in O(2)$. Thus g is a reflection across a line x radians off the x axis and $g = \begin{bmatrix} \cos(2x) & \sin(2x) \\ \sin(2x) & -\cos(2x) \end{bmatrix}$. $R_\theta g^{-1} = R_\theta g^T = \begin{bmatrix} \cos(2x+y) & \sin(2x-y) \\ \sin(2x+y) & -\cos(2x-y) \end{bmatrix}$. $gR_\theta g^T =$
 $\begin{bmatrix} \cos(y) & \sin(y) \\ -\sin(y) & \cos(y) \end{bmatrix} = \begin{bmatrix} \cos(y) & -\sin(-y) \\ \sin(-y) & \cos(y) \end{bmatrix} = \begin{bmatrix} \cos(-y) & -\sin(-y) \\ \sin(-y) & \cos(-y) \end{bmatrix} = R_{-y}$. $\square$